

EDUCATION

University of North Carolina at Chapel Hill

Ph.D. in Biostatistics

Dissertation: Causal Inference Methods for Infectious Disease Prevention Trials

Advised by: Michael Hudgens, Ph.D.

Chapel Hill, NC

Jan 2022–May 2026

University of North Carolina at Chapel Hill

B.S.P.H. in Biostatistics, B.A. in Mathematics

Chapel Hill, NC

Aug 2018–Dec 2021

PROFESSIONAL EXPERIENCE

Wake Forest University

Assistant Professor, Department of Statistical Sciences

Winston Salem, NC

Jul 2026–Present

University of North Carolina at Chapel Hill

Graduate Research Assistant, Department of Biostatistics, Interference Lab

Mentor: Michael Hudgens, Ph.D.

Chapel Hill, NC

Aug 2024–May 2026

University of North Carolina at Chapel Hill

Graduate Research Assistant, Department of Biostatistics, Garcia Lab

Mentor: Tanya Garcia, Ph.D.

Chapel Hill, NC

May 2022–Aug 2024

University of North Carolina at Chapel Hill

Graduate Research Assistant, Department of Biostatistics, Center for AIDS Research

Mentor: Michael Hudgens, Ph.D.

Chapel Hill, NC

Jan 2022–May 2024

National Cancer Institute, National Institutes of Health

Research Intern, Biostatistics Branch, Division of Cancer Epidemiology and Genetics

Mentor: Jianxin Shi, Ph.D.

Bethesda, MD

Summer 2021

University of North Carolina at Chapel Hill

Research Assistant, Department of Psychiatry

Principal Investigator: Susan Girdler, Ph.D.

Chapel Hill, NC

Jul 2017–Jul 2020

TEACHING EXPERIENCE

- **Instructor** at Wake Forest University Fall 2026
Elementary Probability and Statistics (STA 111)
- **Instructor** at University of North Carolina at Chapel Hill Fall 2024, Fall 2025
Topics in Real Analysis (BIOS 672)
- **Instructor** at University of North Carolina at Chapel Hill Spring 2025
Principles of Statistical Inference (BIOS 600)
- **Teaching Assistant** at University of North Carolina at Chapel Hill Fall 2023
Introduction to Biostatistics (BIOS 500H)
- **Teaching Assistant** at University of North Carolina at Chapel Hill Spring 2023
Design of Public Health Studies (BIOS 668)

- **Tutor** at University of North Carolina at Chapel Hill Fall 2022
Calculus of Functions of One Variable I (MATH 231)
Masters Level Probability and Statistical Inference I (BIOS 660)
- **Tutor** at Chapel Hill High School Aug 2020–May 2022
Algebra, Geometry, Pre-Calculus, AP Calculus BC

PUBLICATIONS

*: Co-first authorship

- [1] S. Lee*, **B.D. Richardson***, Y. Ma, K. S. Marder, and T. P. Garcia, “Sparcc: Semi-parametric robust estimation in a right-censored covariate model”, *Journal of the American Statistical Association*, 2025+.
- [2] **B.D. Richardson***, B. Blette*, P. Gilbert, and M. Hudgens, “Addressing confounding and continuous exposure measurement error using corrected score functions”, en, *Biometrics*, vol. 81, no. 2, Apr. 2025, Publisher: Oxford University Press (OUP), ISSN: 0006-341X, 1541-0420.
- [3] B. Chi, F. Saidi, L. Graybill, T. Phanga, K. Mollan, K. Amico, K. Freeborn, N. Rosenberg, L. Hill, T. Hamoonga, **B.D. Richardson**, T. Kalua, S. Phiri, and W. Mutale, “A patient-centered, combination intervention to support adherence to HIV pre-exposure prophylaxis during pregnancy and breastfeeding: A randomized pilot study in Malawi”, *JAIDS Journal of Acquired Immune Deficiency Syndromes*, vol. 95, pp. 42–51, 2024.
- [4] L. A. Graybill, B. H. Chi, T. E. Hamoonga, M. Kasaro, J. N. Hodges, **B.D. Richardson**, J. S. Bissram, F. Saidi, K. R. Mollan, K. Freeborn, N. E. Rosenberg, K. A. Powers, and W. Mutale, “Predictors of maternal HIV acquisition during pregnancy and lactation in sub-Saharan Africa: A systematic review and narrative synthesis”, en, *PLOS ONE*, vol. 19, no. 12, G. T. Feyissa, Ed., e0314747, Dec. 2024, ISSN: 1932-6203.
- [5] S. C. Lotspeich, M. C. Ashner, J. E. Vazquez, **B.D. Richardson**, K. F. Grosser, B. E. Bodek, and T. P. Garcia, “Making sense of censored covariates: Statistical methods for studies of Huntington’s disease”, en, *Annual Review of Statistics and Its Application*, vol. 11, no. 1, pp. 255–277, Apr. 2024, ISSN: 2326-8298, 2326-831X.
- [6] S. Lotspeich*, **B.D. Richardson***, P. Baldoni, K. Enders, and M. Hudgens, “Quantifying the HIV reservoir with dilution assays and deep viral sequencing”, *Biometrics*, 2024.
- [7] J. J. Peterson, C. A. Lewis, S. D. Burgos, A. Manickam, Y. Xu, A. A. Rowley, G. Clutton, **B.D. Richardson**, F. Zou, J. M. Simon, D. M. Margolis, N. Goonetilleke, and E. P. Browne, “A histone deacetylase network regulates epigenetic reprogramming and viral silencing in HIV-infected cells”, en, *Cell Chemical Biology*, vol. 30, no. 12, 1617–1633.e9, Dec. 2023, ISSN: 24519456.
- [8] J. Shi, ..., **B.D. Richardson**, ..., and Q. Lan, “Genome-wide association study of lung adenocarcinoma in east asia and comparison with a European population”, *Nature Communications*, vol. 14, p. 3043, 2023.

SERVICE

- **MAPS Director** at University of North Carolina at Chapel Hill May 2023–Present
Director of Mentorship and Advice for Prospective Students (MAPS) program
- **MAPS Mentor** at University of North Carolina at Chapel Hill Aug 2022–Present
Mentor for prospective students with MAPS program
- **Session Chair** “Statistical Challenges with Linked and Coarsened Data in Public Health Research” Mar 2024
ENAR Spring Meeting, Baltimore, MD

PRESENTATIONS

- **“Social Distancing to Reduce Transmission of Influenza-Like-Illness on College Campuses: UNC Biostatistics Recruiting Event”** Feb 2026
Invited talk, Chapel Hill, NC
- **“Social Distancing to Reduce Transmission of Influenza-Like-Illness on College Campuses: Wake Forest Statistical Sciences Colloquium”** Dec 2025
Colloquium, Winston-Salem, NC
- **“Causal Inference for Infectious Disease Prevention: Bowdoin Undergraduate Colloquium”** Nov 2025
Colloquium, Brunswick, ME
- **“Addressing Confounding and Continuous Exposure Measurement Error Using Corrected Score Functions”** Aug 2025
Invited talk, JSM, Nashville, TN
- **“When the Life Sciences Give You Lemons: Making the Most of Coarsened Data in Public Health Research”** Feb 2025
Invited Talk, UNC Department of Biostatistics, Chapel Hill, NC
- **“Addressing Confounding and Continuous Exposure Measurement Error Using Corrected Score Functions”** Jan 2025
Seminar, UNC Causal Inference Research Lab, Chapel Hill, NC
- **“Robust and efficient estimation in the presence of a randomly censored covariate”** Nov 2024
Invited talk, UNC Biostatistics 75th Anniversary, Chapel Hill, NC
- **“Quantifying the HIV Reservoir With Dilution Assays and Deep Viral Sequencing”** Oct 2024
Invited talk, UNC BIOS Student Seminar, Chapel Hill, NC
- **“Robust and efficient estimation in the presence of a randomly censored covariate”** Aug 2024
Contributed talk, JSM, Portland, OR
- **“Quantifying the HIV Reservoir With Dilution Assays and Deep Viral Sequencing”** Apr 2024
Invited talk, UNC BIOS/STOR Joint Student Seminar, Chapel Hill, NC
- **“Doubly robust estimation under a randomly censored covariate”** Mar 2024
Contributed talk, ENAR Spring Meeting, Baltimore, MD
- **“Quantifying the HIV Reservoir With Dilution Assays and Deep Viral Sequencing”** Mar 2023
Contributed talk, ENAR Spring Meeting, Nashville, TN
- **“Crash Course: Biostatistics and Epidemiology”** May 2022
Invited lecture, ENABLE HiDAV Boot Camp, University of North Carolina at Chapel Hill

SKILLS

- **Statistical Computing:** R, SAS, SQL
- **Languages:** English (native), Spanish (proficient)

SCHOLARSHIPS AND AWARDS

- Inductee, Delta Omega Honorary Society in Public Health 2026
- Spirit of Service Award, UNC Department of Biostatistics 2026
- Dennis Gillings Innovation Award, UNC Department of Biostatistics 2026
- Service Award, NC Chapter of the American Statistical Association 2026
- T32 Predoctoral Traineeship, National Institute of Environmental Health Sciences 2022–2025